Camilo A. F. Salvador

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SUMMARY

D.Eng. researcher working at the interface of materials science, physics, and data science. My current research focuses on coupling machine learning with high-throughput characterization techniques (in situ TEM, 4D-STEM, high-energy diffraction, APT) to design new structural materials for the automotive, biomedical, and energy sectors. I have a particular interest in techniques that involve latent representations of characterization and atomistic data, including transformers, diffusion probabilistic models, and foundation models such as JEPA. I have published 28 papers, with 868 citations and an h-index of 17. For an updated list of publications, please refer to Google Scholar.

RESEARCH EXPERIENCE

Research Engineer (Contract) 2024 – 2027
Université Paris-Saclay, CEA, SRMP *Gif-Sur-Yvette, France*
In situ TEM ion-irradiation at the JaNNuS-Orsay platform; deep-learning-based post-processing of TEM data with detectors, segmentation models, and transformers; large-scale molecular dynamics (MD) simulations to mimic damage accumulation under irradiation; development and use of HPC/GPU code to generate atomistic descriptors.

Postdoc 2022 – 2024
CNRS, SIMaP, Grenoble-INP *Grenoble, France*
Manufactured and characterized stainless-steel diffusion couples. Performed high-throughput experiments at the ESRF; developed a small-angle scattering (SAS) modeling code; 3-month internship at the Max Planck Institute for Sustainable Materials (Düsseldorf) covering FIB needle preparation, APT operation, and post-analysis with IVAS and custom Python codes.

Postdoc 2019 – 2021
University of São Paulo *São Paulo, SP, Brazil*
Applied first-principles calculations (DFT) and machine learning to model the elastic behavior of multicomponent alloys; used CNNs to interpret confined fluid flow paired with the lattice Boltzmann method.

Doctor of Engineering 2015 – 2019
University of Campinas *Campinas, SP, Brazil*
Thesis on Ti–Nb–Fe and Ti–Nb–Fe–Zr alloys for orthopedic implants; combined phase-transformation and diffusion modeling with microstructural and mechanical characterization to study 17 compositions, identifying two alloys with about 30% higher strength than the Ti–Nb–Ta–Zr benchmark.

Research internship 2017 – 2018
OSU, The Ohio State University *Columbus, OH, USA*
Microstructural optimization of aeronautical and aerospace Ti-alloys; prepared FIB lamellae and performed TEM/STEM analyses at CEMAS under Prof. Hamish Fraser.

EDUCATION

University of Campinas Campinas, SP, Brazil
M.S. in Mechanical Engineering, Materials Science *2012 – 2015*

Federal University of ABC Santo André, SP, Brazil
Double degree: B.S. in Science & Technology, Engineer diploma (Materials) *2008 – 2012*

SKILLS

Specialized techniques: Transmission electron microscopy (TEM, 2,000+ hours), focused-ion beam (FIB, 400+ hours), synchrotron X-ray or neutron diffraction (5 projects across 3 facilities, 400+ hours), atom-probe tomography (APT, 2 projects, 100+ hours)

Languages: Portuguese (Native), English (C2), French (B2), Spanish (B1)

Programming: Python, MATLAB, C++, R, CUDA, Fortran

Deep Learning Frameworks: PyTorch, TensorFlow, Keras

Others: NumPy, Pandas, SciPy, OpenCV, Git, Slurm, LaTeX